# ASSIGNMENT COVERSHEET

**UTS: ENGINEERING & INFORMATION TECHNOLOGY**

| SUBJECT NUMBER & NAME | NAME OF STUDENT(s) (PRINT CLEARLY) | STUDENT ID(s) |
|---|---|---|
| 32146 Data Visualisation and Visual Analytics | *Srichairattanakul* (SURNAME) *Warit* (FIRST NAME) | 25745973 |

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| NAME OF TUTOR | TUTORIAL GROUP | DUE DATE |
|---|---|---|
| Sylvan Rudduck | 4 | 9 September |

**ASSESSMENT ITEM NUMBER & TITLE**

Assessment Task 3: Visual Analytics

☐ I acknowledge that if AI or another non-recoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used.
☐ I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.
☐ I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.
☐ I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

**Declaration of originality**: The work contained in this assignment, other than that specifically attributed to another source, is that of the author(s) and has not been previously submitted for assessment. I have rewritten any material provided by AI or other nonrecoverable sources and where appropriate acknowledged their contribution. I understand that, should this declaration be found to be false, disciplinary action could be taken and penalties imposed in accordance with University policy and rules. In the statement below, I have indicated the extent to which I have collaborated with others, whom I have named.

No content generated by AI technologies or other sources has been presented as my own work and I have rewritten any text provided by AI or other sources in my own words.

**Statement of collaboration**:

**Signature of student(s) ____Warit Srichairattanakul_____ Date ___8 November 2025_____**

# ASSIGNMENT RECEIPT

To be completed by the student if a receipt is required

| SUBJECT NUMBER & NAME | NAME OF TUTOR |
|---|---|
| | |

| SIGNATURE OF TUTOR | RECEIVED DATE |
|---|---|
| | |

# STYLE GUIDE for ASSIGNMENT SUBMISSION

Before submitting an assignment, you should refer to the policies and guidelines set out in the following:

- FEIT Student Guide
- UTS Library - referencing
- HELPS - English and academic literacy support
- UTS GSU - coursework assessment policy and procedures

Unless your Subject Coordinator has indicated otherwise in the Subject Outline, you must follow the instructions below for submission of assignments in the Faculty of Engineering and Information Technology.

## Writing style

It is usually best to write your initial draft in the default settings of your software without formatting. Use the following guides in your writing.

**Purpose and audience**: use the correct genre and language style expected for the particular task.

**Language**: use 'plain English' for all technical writing. More information about this language style can be found at www.plainenglish.co.uk/free-guides.html.

Use spelling and grammar software tools to check your writing. Edit your document.

**Standards**: always use:

- Australian spelling standards (Macquarie Dictionary)
- SI (International System of Units) units of measurement
- ISO (International Organisation for Standardisation) for writing dates and times for international documents. For example **yyyy-mm-dd** or **hh-mm-ss**. However, for most applications it is more helpful to present the date in full as **26 August 2016**.

**Graphics and tables** should:

- be numbered
- have an appropriate heading and/or caption
- be fully labelled
- be correctly referenced.

## Presentation

Unless otherwise instructed, all assignment submissions should be **word processed** using spell-check and grammar-check software. Work should be well **edited** before submission. Use the following default settings:

**Page setup**: set margins at no less than 20mm all around.

**Paper**: print on A4 bond, double-spaced and preferably double-sided, left justified.

**Font**: use the software default style to provide consistency. The recommended style includes:

- 10-12 pt font
- consistent formatting with a limited number of fonts
- lines no more than 60 characters (use wider margins or columns if you need to make lines shorter)

**Header** should include:

- your name and student number
- the title of the paper or task.

**Footer** should include the page number and current date.

Cover sheet and statement of originality: all work submitted for assessment must be the original work of the student(s) submitting the work. A standard faculty cover sheet (see over) must be attached to the front of the submission. Any collaboration between the submitting student and others must be declared on the cover sheet.

## Referencing

All sources of information used in the preparation of your submission must be acknowledged using the APA system of referencing. This includes all print, video, electronic sources.

Phrases, sentences or paragraphs taken verbatim from a source must be in quotation marks and the source(s) cited using both **in-text** referencing and a **reference list**.

Plagiarism is the failure to acknowledge sources of information. You should be fully aware of the meaning of plagiarism and its consequences both to your marks, position at the university and criminal liability. The plagiarism in your assignment submissions can be assessed both in hard copy and in soft copy through software such as Turnitin.

The UTS Library and UTS HELPS (web links above) provide extensive information for students on referencing correctly to support you in avoiding plagiarism.

# Executive Summary

This report presents a comprehensive visual analysis of Australia's international trade between 1988 and 2024, focusing on both import and export patterns. The project aims to uncover analytical and statistical trends across all main categories, with a deeper exploration of the Machinery and Transport Equipment sector, the largest contributor to Australia's trade activity. Using Tableau dashboards and storyboards, various visualisation techniques, such as line charts, stacked area graphs, and treemaps, were applied to highlight major trade movements, category relationships, and key turning points throughout the study period.

The findings reveal that Australia's trade has expanded significantly over the past three decades, with strong export growth after 2020 and continued dominance of machinery-related imports. Within this category, Road Vehicles emerged as the leading subcategory, reflecting the country's heavy dependence on imported automotive products and industrial equipment. Forecast analysis from 2015 to 2029 suggests that this reliance will persist, although opportunities exist for growth in domestic manufacturing and electric vehicle (EV) production.

By integrating dashboards and storyboards, this study not only identifies long-term economic patterns but also provides an intuitive narrative for understanding the causes behind them. The combination of analytical precision and storytelling clarity makes this approach highly effective for visualising complex trade data, offering insights that can support evidence-based decision-making and future policy planning for sustainable trade development.

# Data Preparation

The Statistical Pattern (Ratios)

The statistical pattern was designed to identify the proportion of each main and subcategory's import or export value relative to the total trade in a given year.
This calculation converts the raw dollar values into a percentage-based representation, allowing fair comparison across categories and over time.

To construct the statistical pattern, the total trade value for each year (column "-1 TOTAL") was used as the denominator, while each category's trade value (e.g., Food and live animals, Machinery and transport equipment) was used as the numerator.
The calculation for each cell is defined as follows:

$$Statistical\ Pattern = \frac{CategoryValue_{i}}{TotalValue_{year}}$$

All results were expressed as decimals (ranging from 0 to 1), representing each category's share of total imports or exports. For instance, a value of 0.185 indicates that the category accounted for 18.5% of total trade in that year. This transformation enabled the dataset to be standardised across years, supporting the creation of stacked area charts and composition trend analyses in Tableau.

The Analytical Pattern (Changes)

The analytical pattern focuses on the year-over-year (YoY) change ratio of import and export values, showing how each category's trade performance fluctuates annually. This method highlights significant shifts, such as the effects of the Global Financial Crisis (2008–2009) and the post-COVID recovery (2021–2023).

The formula used compares the trade value of each year with the previous year's value:

$$Analytical\ Pattern = \frac{Value_{t}}{Value_{t-1}}$$

Where:

$Value_{t}$ represents the trade value (import or export) for a specific category in the current year.

$Value_{t-1}$ represents the corresponding value from the previous year.

A result greater than 1 indicates growth, while a value below 1 represents a decline compared with the previous year. For example, a ratio of 1.15 means the category's trade increased by 15%, whereas a ratio of 0.90 indicates a 10% decrease. This calculation was applied to every category and subcategory separately for both imports and exports. The transformed data was then used to create line graphs in Tableau to visualise temporal growth trends and highlight years of economic disruption or recovery.

# Visualisations

1. Statistical Pattern of Main Categories in Time-Series

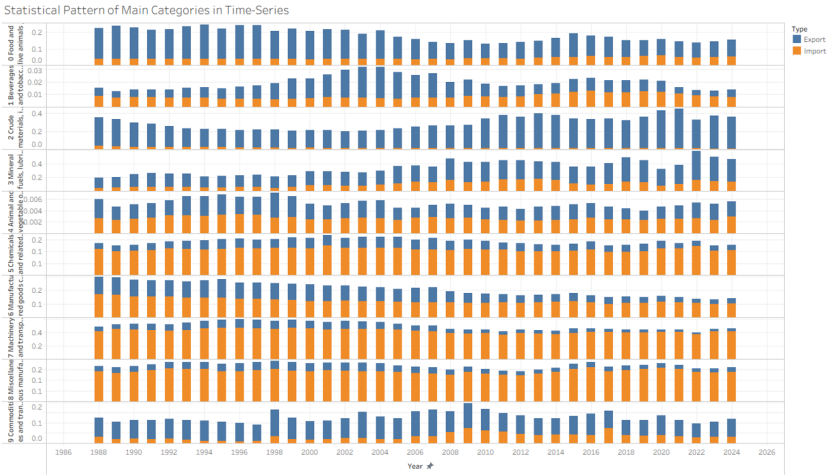



Figure 1. Statistical Pattern of Australia's Main Import and Export Categories (1988-2024)

Figure 1 illustrates how the main import and export categories of Australia have changed in proportion from 1988 to 2024. A stacked bar chart was used to compare the percentage share of each category in total trade each year. The orange bars represent imports, while the blue bars show exports.

From the chart, we can see that Machinery and Transport Equipment takes up the biggest share in both imports and exports every year. This means Australia relies a lot on machines, vehicles, and industrial equipment from other countries. Food, live Animals, and Manufactured Goods stay quite stable over time, showing that these sectors keep a steady contribution to trade.

Around 2008 to 2009, there was a clear drop in the share of Mineral Fuels and Crude Materials, which was linked to the global financial crisis when global demand fell. After 2020, the import share of machinery rises again, likely because of infrastructure projects and the growing use of electric vehicles after COVID-19.

2. Analytical Pattern of Main Categories in Time-Series

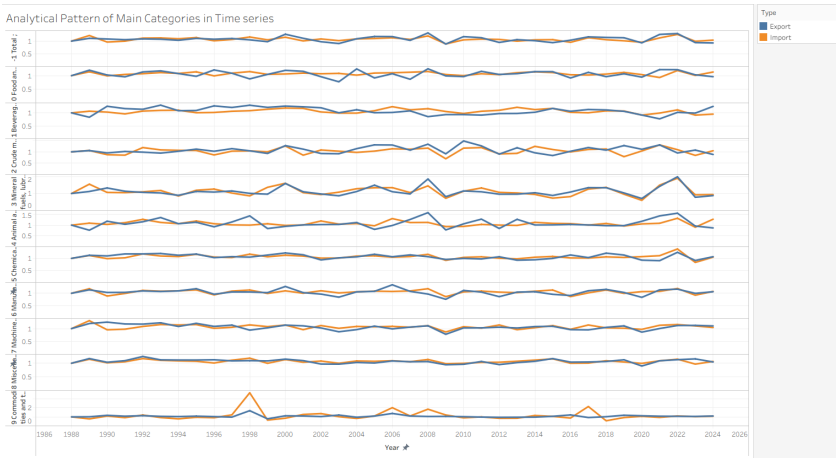



Figure 2. Analytical Pattern of Australia's Main Import and Export Categories (1988-2024)

Figure 2 presents the analytical pattern of Australia's main import and export categories from 1988 to 2024. A line chart was used to show the year-over-year changes in trade performance for each main category. The blue lines represent export trends, while the orange lines represent imports.

From the visualisation, most categories show consistent movement over time, with exports and imports generally following similar directions. Machinery and Transport Equipment remains one of the

most stable yet gradually increasing categories, highlighting Australia's long-term dependency on machinery and vehicles. Meanwhile, Mineral Fuels and Crude Materials show more visible fluctuations, especially around 2008-2009, when the global financial crisis affected international trade and commodity prices.

In the years after 2020, small upward trends can be seen in machinery imports, which might relate to infrastructure investment and the adoption of electric vehicles. Overall, the chart indicates that although Australia's trade patterns fluctuate with global events, the key industrial sectors continue to show resilience and gradual recovery over time.

3. Import and Export of Main Categories in Time Series

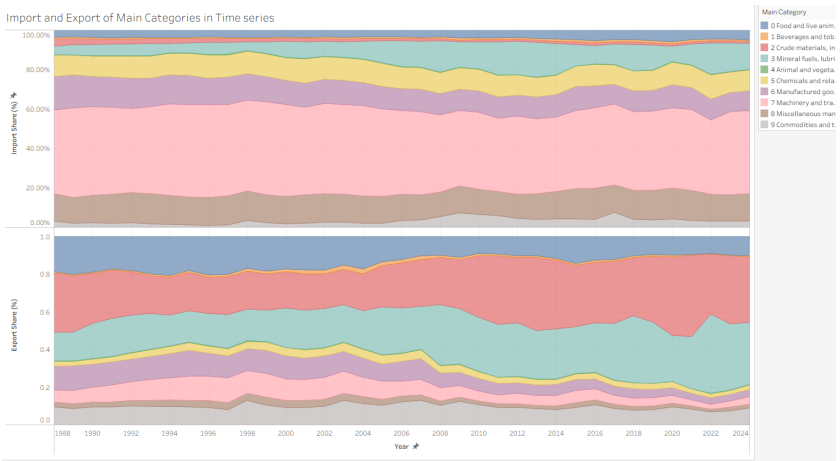



Figure 3. Import and Export Share of Main Categories in Time-Series (1988-2024)

Figure 3 visualises the overall structure of Australia's trade by showing the proportion of each main category in total imports and exports from 1988 to 2024. The stacked area chart allows us to easily compare how different sectors contribute to trade over time, with the upper half representing import shares and the lower half representing exports.

From the visualisation, Machinery and Transport Equipment again stands out as the dominant category in imports, maintaining a large share across the years. This reflects Australia's dependence on industrial machinery, vehicles, and technology from other countries. On the export side, Crude Materials and Mineral Fuels occupy the largest portions, highlighting the country's strength in resource-based industries such as mining and energy.

Between 2008 and 2009, a visible contraction in the export share of *Mineral Fuels* aligns with the global financial crisis that affected global demand. However, after 2020, the overall export structure became more stable again, showing a gradual recovery driven by strong global demand for energy and raw materials.

4. Tree Map of Main Categories

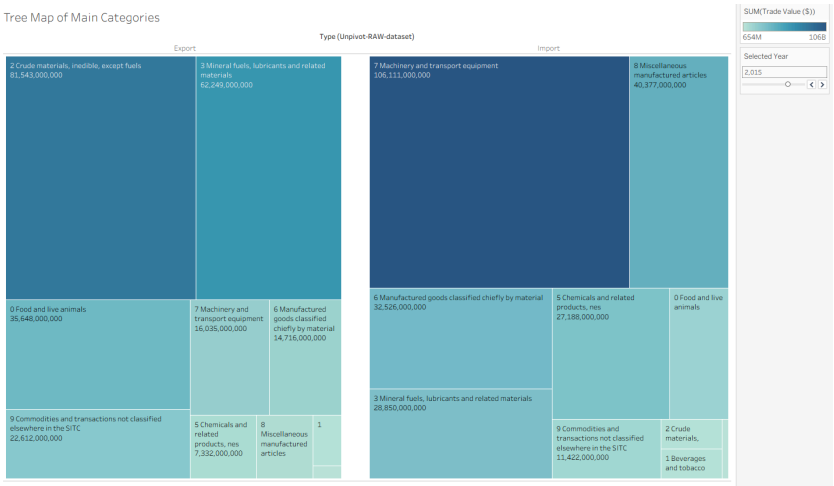



Figure 4. Tree Map of Main Categories

Figure 4 presents a tree map that compares the main import and export categories of Australia in 2015. Each rectangle represents one trade category, and its size corresponds to the total trade value, making it easy to compare which categories dominate Australia's trade structure. The left side shows exports, while the right side shows imports.

This dashboard also includes an interactive year slider, allowing users to adjust the year and instantly observe how the trade composition changes over time. By moving the slider, users can explore shifts in dominant categories and understand how Australia's trade focus evolved from raw materials toward more industrial and manufactured products.

From the visualisation, Crude Materials and Mineral Fuels are the largest export categories, reflecting Australia's strong role as a global supplier of raw materials and energy resources. Meanwhile, Machinery and Transport Equipment stands out as the largest import category, valued at over $106 billion, showing Australia's high dependence on imported machinery, vehicles, and technology equipment.

Other major import groups, such as Manufactured Goods and Chemicals, also play important roles in supporting the industrial and construction sectors. On the export side, categories like Food and Live Animals and Manufactured Goods contribute smaller but consistent shares, representing the diversification of Australia's trade portfolio.

5. Import vs Export of Machinery and Transport Equipment

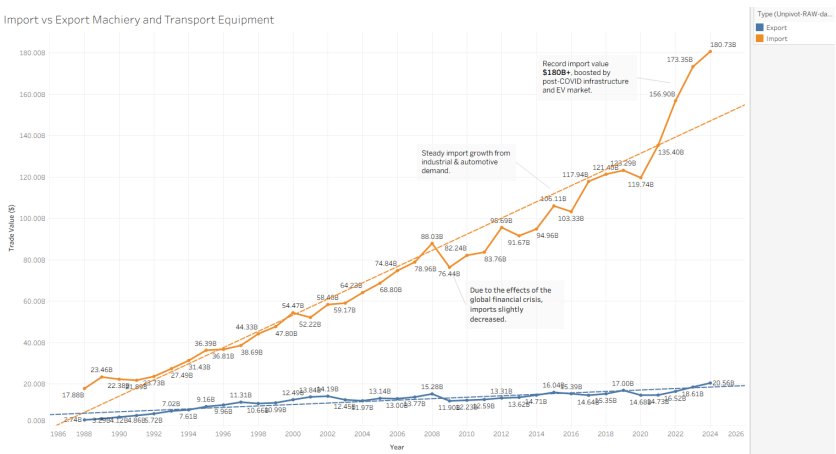


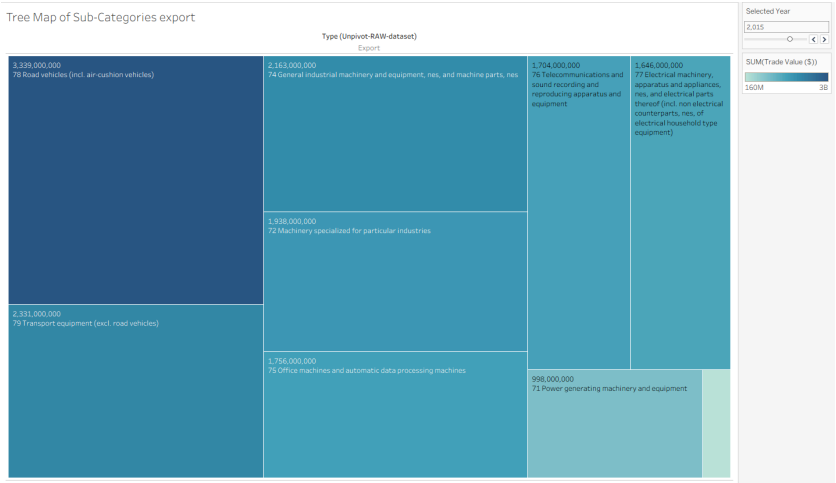

Figure 5. Import vs Export of Machinery and Transport Equipment (1988-2024)

Figure 5 shows the long-term trend of Australia's import and export values in the Machinery and Transport Equipment category between 1988 and 2024. The orange line represents imports, while the blue line shows exports. The dotted trendlines help visualise the overall direction of trade growth over time.

From this chart, it is clear that imports in this category have grown significantly faster than exports. The steady increase since the late 1990s reflects Australia's strong industrial and automotive demand, particularly for equipment, vehicles, and electrical machinery. Imports dipped slightly during the 2008-2009 global financial crisis but quickly recovered, following the pattern of overall economic rebound.

A sharp rise appears after 2020, with import values surpassing $180 billion in 2024, the highest in the time series. This increase aligns with post-COVID infrastructure investment and the expansion of the electric vehicle (EV) market, both of which require substantial machinery and technological components from abroad. In contrast, exports in this category remain relatively low throughout the period, showing that Australia's machinery sector is much more dependent on imports than on export capacity. The consistent gap between the two lines emphasises the country's structural reliance on high-value industrial goods produced overseas.

6. Export of Machinery and Transport Equipment Sub-Categories




Tree Map of Export Machinery and Transport Equipment Sub-Categories

Figure 6 presents a tree map that breaks down the export values of Australia's Machinery and Transport Equipment category in 2015 into its sub-categories. Each block represents one sub-category, and the size of the block indicates its contribution to the total export value of this main category. The chart also includes an interactive year slider, which allows users to switch between years and observe how each sub-category's export share changes over time.

From the visualisation, Road Vehicles stand out as the largest export group, valued at around $3.3 billion, followed by Transport Equipment (excluding road vehicles) and General Industrial Machinery. These three categories together account for the majority of machinery exports. The dominance of vehicles and industrial equipment shows that even though Australia mainly imports high-value machines, it still plays a role in exporting specific machinery types, particularly vehicles and parts related to transport industries. Other categories, such as Telecommunications Equipment and Electrical Machinery, also contribute significantly, reflecting the growing influence of the electronics and communication sector in 2015.

7. Import of Machinery and Transport Equipment Sub-Categories

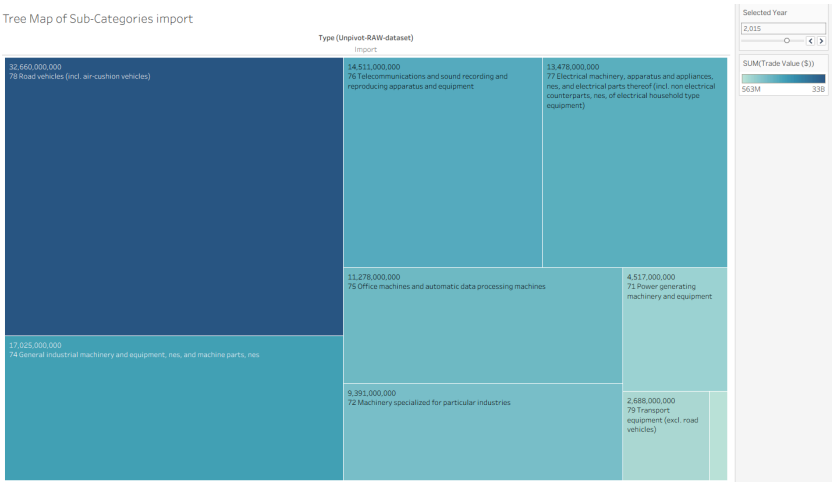



Figure 7. Tree Map of Import Machinery and Transport Equipment Sub-Categories

Figure 7 illustrates the import values of Australia's Machinery and Transport Equipment category in 2015, broken down into detailed sub-categories. Similar to the previous figure, this chart uses a tree map to visualise how each sub-sector contributes to total imports. Each rectangle's size represents its trade value, while colour shading indicates the relative weight within the category. Users can also adjust the interactive year slider on the right-hand side to explore changes over time.

From the chart, Road Vehicles clearly dominate as the largest import group, worth over $32 billion, showing Australia's heavy reliance on imported cars, trucks, and other motor vehicles. This reflects the country's limited domestic car manufacturing and strong consumer demand for foreign automobile brands.

Following this, Telecommunication Equipment, Electrical Machinery, and General Industrial Machinery are also major import sectors, indicating the growing influence of technology, communication, and industrial automation in Australia's economy. Imports in these areas are essential to support national infrastructure, digital connectivity, and manufacturing productivity.

By comparison, Power Generating Machinery and Transport Equipment, excluding road vehicles, take smaller shares, but they remain vital for the energy and logistics sectors.

8. Import vs Export of Machinery and Transport Equipment

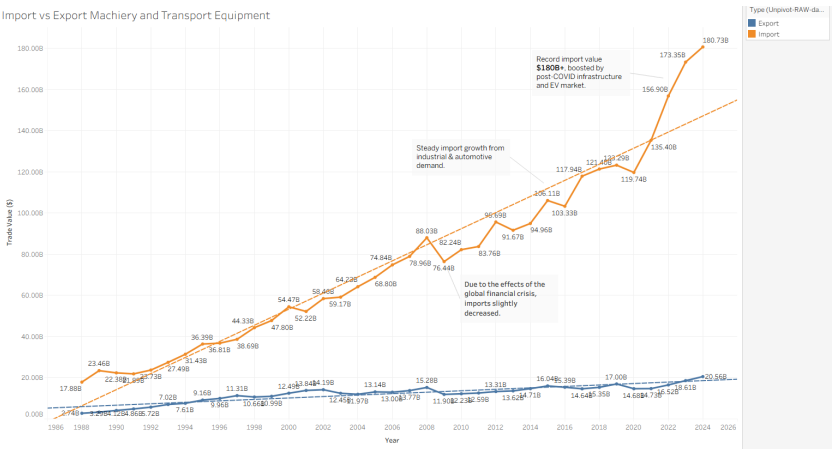



Figure 8. Import vs Export of Machinery and Transport Equipment (1988-2024)

Figure 8 compares the import and export values of Australia's Machinery and Transport Equipment category from 1988 to 2024. The orange line represents import values, while the blue line represents exports. Both are plotted as time-series data to highlight long-term changes in trade performance.

From the chart, imports show a strong and steady upward trend, driven by increasing industrial, infrastructure, and automotive demand. The early growth during the 1990s reflects Australia's industrial expansion and reliance on foreign technology and machinery. Although a small decline appeared around 2008-2009 due to the global financial crisis, imports quickly recovered in the following years.

After 2020, import values rose sharply, reaching over $180 billion in 2024, the highest level in the dataset. This rapid growth was likely influenced by post-COVID recovery projects, new infrastructure investments, and the rising adoption of electric vehicles (EVs), which required more imported machinery and components.

In contrast, exports in this category remain relatively flat over the entire period, rarely exceeding $20 billion. This wide gap between import and export lines highlights a clear trade imbalance in Australia's machinery sector. It shows that the country depends heavily on imported equipment and technology rather than producing and exporting these goods itself.

9. Comparison of Machinery and Transport Equipment Sub-Categories

Figure 9. Export of Machinery and Transport Equipment Sub-Categories

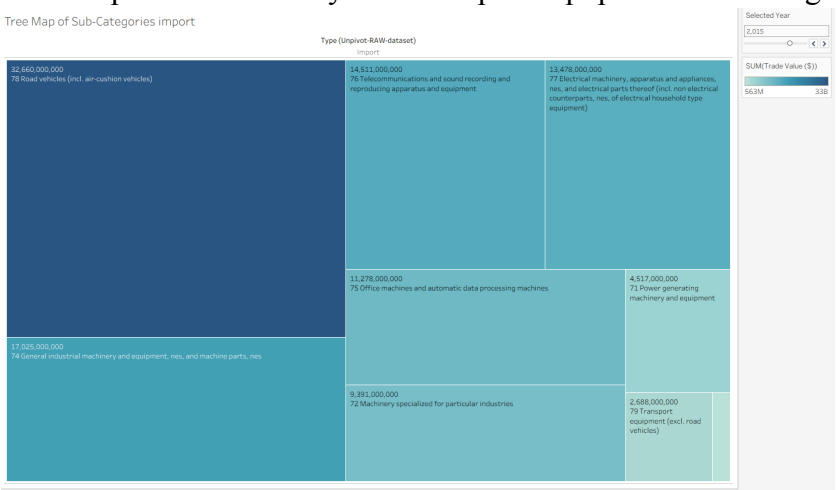



Figure 10. Import of Machinery and Transport Equipment Sub-Categories

Figures 9 and 10 present a pair of tree maps comparing Australia's Machinery and Transport Equipment sub-categories for imports and exports in 2015. Each block represents one sub-category, with its size showing the trade value and its shade indicating relative weight within the category. An interactive year slider is included on the right-hand side, allowing users to change the year and see how each group's value shifts over time.

From both visuals, Road Vehicles dominate on both import and export sides. However, the difference in value is striking; imports reached about $32.6 billion, while exports were only around $3.3 billion. This gap highlights Australia's high dependency on imported vehicles and its limited local car manufacturing capability. Other important import sub-categories include Telecommunication Equipment, Electrical Machinery, and General Industrial Machinery, all showing strong demand in technology,

electronics, and industrial equipment. In contrast, exports are more diversified but smaller in value, led by Transport Equipment and Industrial Machinery.

10. Machinery and Transport Equipment Import and Export by Subcategory

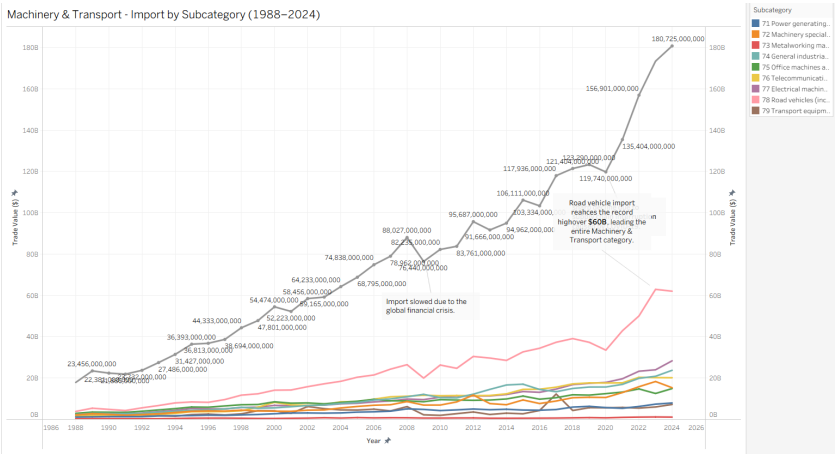



Figure 11. Import of Machinery & Transport subcategory

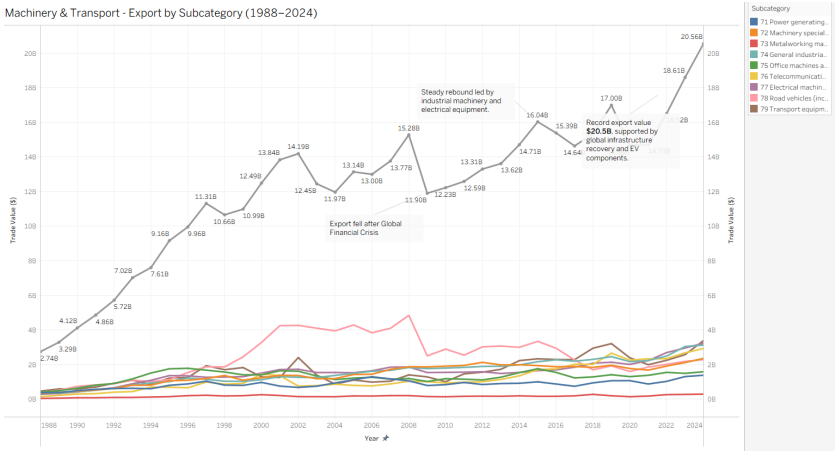



Figure 12. Export of Machinery & Transport subcategory

Figures 11 and 12 show the long-term trends of Australia's Machinery and Transport Equipment imports and exports from 1988 to 2024. The charts reveal that Road Vehicles dominate imports, rising sharply from around $23 billion in 1988 to over $180 billion in 2024. This growth reflects strong demand for vehicles and industrial equipment, boosted further by post-COVID infrastructure projects and the EV market. Other import subcategories, such as Electrical Machinery and Telecommunication Equipment, also show steady increases, highlighting Australia's reliance on advanced technology and electronics.

Exports, in contrast, remain much lower, peaking at $20.5 billion in 2024 with moderate growth led by Industrial Machinery and Electrical Equipment. Overall, these trends show a clear trade imbalance: Australia continues to depend heavily on imported machinery and vehicles, while its exports in this sector grow only modestly.

# Dashboard

## 1. Australia's Trade Overview (1988-2024)

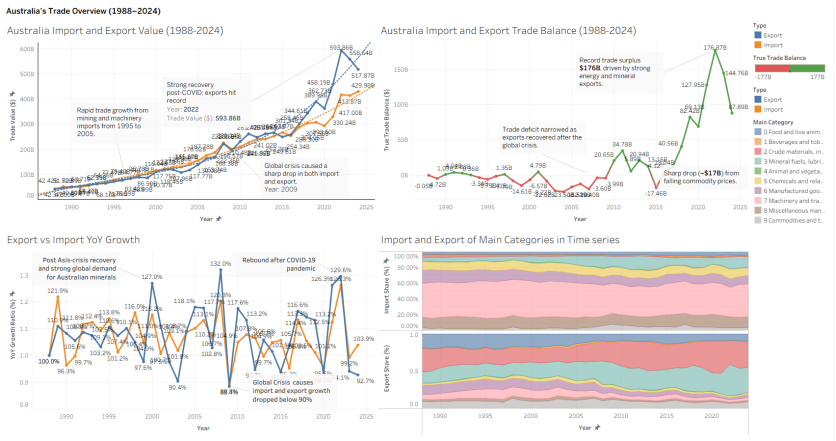



Figure 13. Dashboard of Australia's Trade Overview (1988-2024)

This dashboard provides an overall picture of Australia's international trade from 1988 to 2024. It shows how both import and export values have grown over time, how trade balance fluctuated through global events, and how year-on-year growth rates responded to economic crises and recoveries. The composition chart also reveals which product categories dominate Australia's trade, highlighting the country's reliance on machinery imports and resource-based exports such as minerals and fuels.

## 2. Main Category Trade Overview

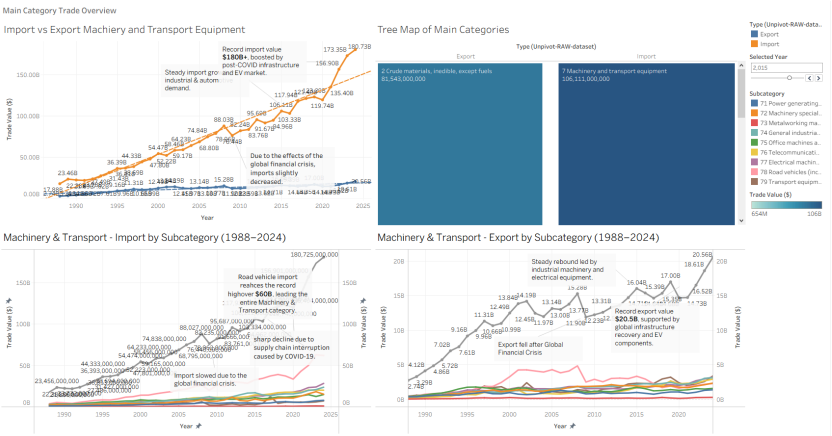



Figure 14. Dashboard of Main Category Trade Overview

This dashboard focuses on Machinery and Transport Equipment, the largest trade category in Australia. It shows how imports and exports in this sector have evolved over time and which subcategories contribute most to trade value.

From the charts, imports have grown strongly since the 1990s, reaching a record of $180 billion in 2024, mainly driven by road vehicles, electrical machinery, and telecommunication equipment. The export value, although smaller, shows steady growth led by industrial machinery and EV-related components after 2020.

The treemap on the right also illustrates that Machinery and Transport Equipment dominates total imports, while Crude Materials lead in exports. Overall, this dashboard highlights Australia's heavy dependence on machinery imports and its limited export capability in the same category.

3. Comparison of Road Vehicles and Transport Equipment Subcategory

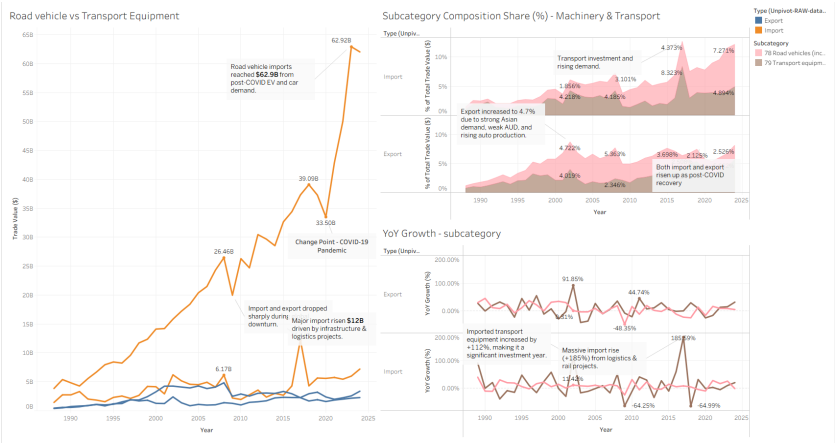



Figure 16. Dashboard of Road Vehicles vs Transport Equipment

This dashboard compares the performance of Road Vehicles and Transport Equipment, the two major subcategories within Australia's Machinery and Transport sector.

The line chart on the left shows that road vehicle imports have risen sharply since 2020, reaching $62.9 billion in 2024, largely driven by EV demand and post-COVID infrastructure recovery. In contrast, transport equipment grew more modestly, with occasional spikes from large-scale logistics and rail projects.

The composition chart on the top right highlights that road vehicles consistently make up the largest share of this category, accounting for over 7% of total trade in recent years. The YoY growth chart at the bottom shows fluctuations linked to major investment years (like +185% growth in 2018) and downturns during crises.

# Storyboard

This storyboard presents a complete storytelling flow of Australia's trade from the overall economy down to the specific subcategory level. It begins by showing Australia's total import and export performance from 1988 to 2024, where both values have risen steadily over time. Major turning points are highlighted, such as the 2009 global financial crisis that caused a temporary drop, and the strong rebound after the COVID-19 pandemic in 2022, which pushed exports to a record high of $593.8B. The trade balance graph further emphasizes Australia's shift from frequent trade deficits before 2015 to significant trade surpluses afterward, mainly due to booming energy and mineral exports.

The analysis then narrows to the main category "Machinery and Transport Equipment", which has shown the most consistent and dominant growth on the import side. The dashboard reveals that Australia heavily relies on this category, with imports reaching around $180B in 2024, supported by infrastructure projects and industrial demand. On the export side, although smaller, this category displays gradual growth led by industrial machinery and electrical equipment, reflecting Australia's participation in global technology and manufacturing chains.

The next part of the storyboard moves deeper into subcategory comparison between Road Vehicles and Transport Equipment. The analysis shows that road vehicles are the key driver of import growth, climbing to $62.9B by 2024, fueled by electric vehicle (EV) adoption and post-pandemic consumer recovery. Meanwhile, transport equipment shows occasional peaks linked to logistics and rail infrastructure investments but remains secondary in value. The year-on-year and composition charts support these insights, showing a rise in the import share of machinery-related products after 2020.

Finally, the forecast visualisation (2015-2029) projects that road vehicle imports will stay high through 2029, while exports remain relatively flat. This suggests that Australia's automotive and machinery sectors will continue to depend heavily on imports rather than domestic production.

Overall, the storyboard clearly connects each level of analysis, from the national trade overview to main category trends and specific subcategory patterns, providing a clear picture of Australia's economic dependency and trade dynamics in both import and export markets.

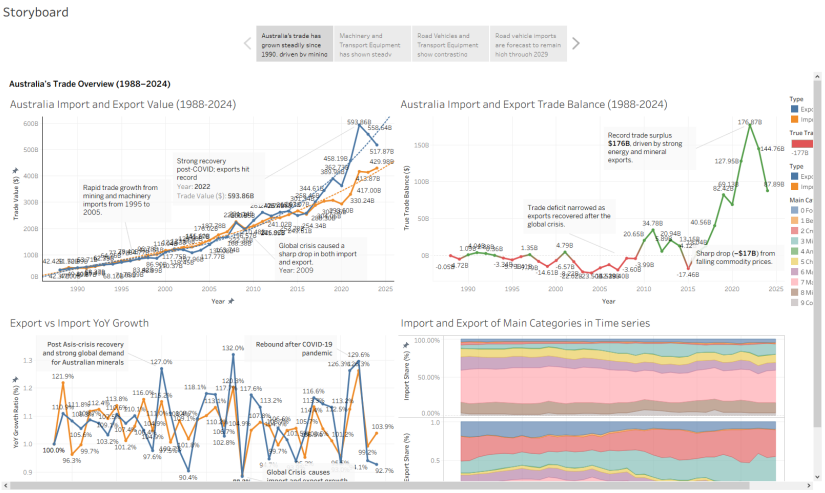



Figure 17. Story point 1 in Storyboard

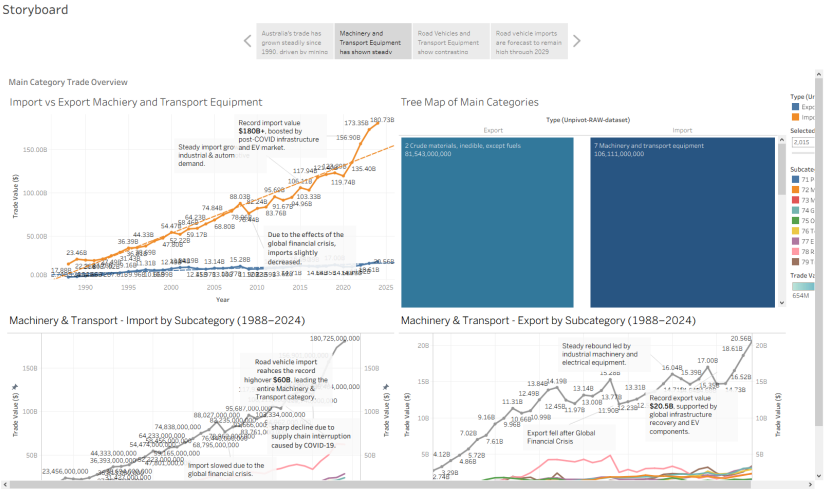



Figure 18. Story point 2 in Storyboard

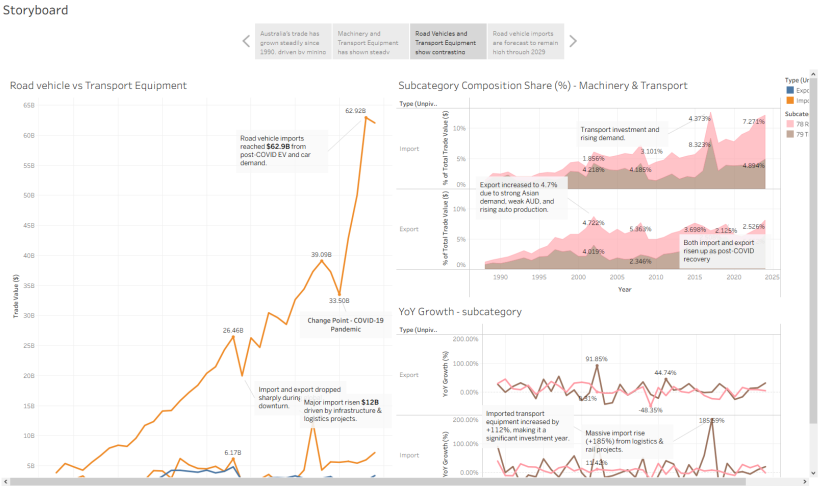



Figure 19. Story point 3 in Storyboard

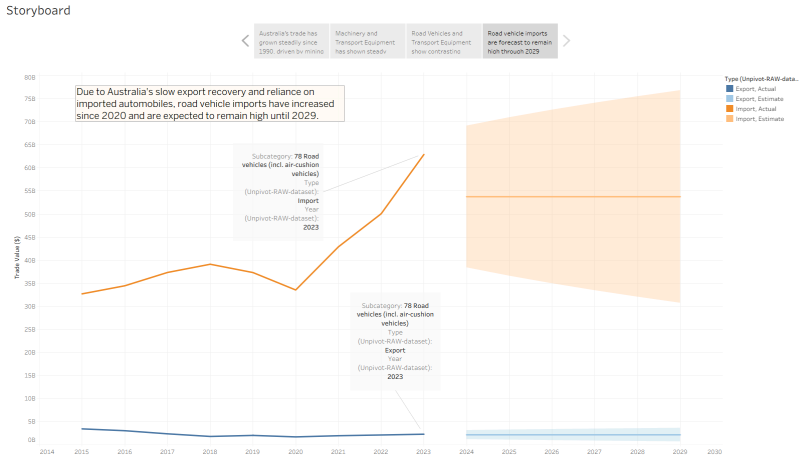



Figure 20. Story point 4 in Storyboard

# Summary of Advantages

The use of both dashboards and storyboards in this analysis supports the understanding of complex trade dynamics through interactive exploration and narrative storytelling.

## Dashboard

The dashboards provide a clear and interactive way to explore complex trade data. They allow users to quickly compare import and export trends across categories, visualize key turning points, and identify dominant sectors like Machinery and Transport Equipment. The combination of line charts, area charts, and tree maps makes it easy to see both the overall picture and detailed breakdowns. Filters and sliders also enhance user interaction, enabling viewers to focus on specific years or categories. Overall, the dashboards transform large datasets into meaningful visual insights that help users understand trade performance more efficiently and intuitively.

## Storyboard

The storyboard effectively connects all dashboards into a cohesive narrative that tells the story of Australia's trade from a broad overview to specific subcategories. It allows viewers to follow a logical flow from total imports and exports, to the main category, and finally to the road vehicle subcategory and its forecast. This structured storytelling approach helps explain not just "what happened" but also "why it happened." By combining visual highlights, captions, and annotations, the storyboard makes the data more engaging and easier to interpret, supporting deeper understanding and clearer communication of economic insights.

# AI Comparison and Product Forecast Analysis: Road Vehicles

The selected product for this part is Road Vehicles, the most dominant subcategory within Machinery and Transport Equipment. Based on the data visualisation in Tableau, imports of road vehicles have steadily increased over the last decade, from approximately $42 billion in 2015 to $62.9 billion in 2024, representing a growth of nearly 49%. This continuous rise is supported by Australia's strong consumer demand, post-COVID infrastructure recovery, and the growing adoption of electric vehicles (EVs). The forecast trend from Tableau predicts that imports will keep rising, reaching around $78 billion by 2029, showing that Australia will likely remain heavily dependent on imported vehicles and automotive technology.

To compare this with an AI analysis, ChatGPT was asked to analyse the same product using the 2015-2024 dataset and to forecast the next five years (2025-2029). ChatGPT suggests that the trend will be upward, increasing about 2.8-3% per year, with total import values expected to reach $76-$79 billion by 2029.

## Advantages and Disadvantages of AI (ChatGPT)

The main advantage of using ChatGPT for analysis is its ability to quickly interpret data and generate statistical forecasts without requiring complex software setup. It provides quantitative insights and textual explanations that complement visual tools like Tableau. However, its limitations include a lack of access to raw numerical computation or real-time datasets and reliance on textual trend estimation rather than actual numerical modelling. While Tableau gives a more data-driven visual understanding, ChatGPT helps explain the underlying logic and broader implications in a more narrative and analytical way.

# Conclusion

This study explored Australia's trade patterns from 1988 to 2024, focusing on the Machinery and Transport Equipment category and its key subcategory, Road Vehicles. Through a combination of statistical and analytical visualisations, the analysis revealed a clear structural imbalance between imports and exports. Imports in this category have grown steadily for more than three decades, reaching over $180 billion in 2024, while exports remain below $21 billion. This gap demonstrates Australia's strong reliance on imported machinery, vehicles, and technology from international markets.

The designed dashboards and storyboards effectively visualised these long-term patterns and major turning points. The Trade Overview Dashboard highlighted key global events such as the 2009 Global Financial Crisis, which caused a sharp drop in trade, and the 2021-2022 post-COVID rebound, which pushed exports to record highs. The Main Category Dashboard then revealed how machinery imports dominate Australia's economy, while Tree Maps and Subcategory Dashboards illustrated the growing importance of Road Vehicles, Electrical Machinery, and Transport Equipment. Together, these visual tools built a clear narrative from the national trade level down to specific industry segments, allowing easier identification of breakthrough trends and economic dependencies.

The forecast analysis extended these insights to 2029. Tableau's predictive model projected that Road Vehicle imports will remain high, rising from $62.9 billion in 2024 to approximately $78 billion by 2029, showing continued dependence on automotive and industrial imports. The AI-based regression analysis by ChatGPT produced a similar pattern, estimating a steady annual growth rate of around 2.8-3.0% and forecasting imports to reach $76-79 billion by 2029. Both methods confirmed the same upward trend, though the AI model provided a narrower confidence range and more precise numerical outputs, while Tableau delivered clearer visual interpretation and contextual understanding.

Based on these findings, several recommendations are proposed for policymakers and industry leaders. Australia should invest in local electric vehicle (EV) assembly and component manufacturing to reduce import dependency and support R&D in industrial machinery to increase export capability. Promoting diversified import sources and technology transfer partnerships would further strengthen trade resilience and competitiveness in the global market.

Overall, this project demonstrates that combining interactive dashboards and AI-assisted analysis provides a powerful method for understanding and forecasting trade dynamics. Dashboards offer analytical depth and visual clarity, while AI enhances precision and provides complementary insights for prediction. Together, they transform complex trade data into meaningful stories that explain past trends, anticipate future outcomes, and support data-driven decision-making for Australia's sustainable industrial growth.